III B.Tech – I Semester					
Course Code	**DESIGN OF REINFORCED**	L	T	P	C
23A01502	**CONCRETE STRUCTURES**	3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental methods of concrete structure design, including elastic, ultimate load, and limit state methods.
2. **Analyze** and**design** reinforced concrete beams, slabs, staircases, columns, and footings using the Limit State Method as per IS codes.
3. **Evaluate** the behavior of reinforced concrete membersin terms of flexure, shear, torsion, bond, and anchorage.
4. **Apply** design principles to ensure serviceability and safety of concrete structures under various loading conditions.
5. **Develop** skills to use design aids and professional software for the analysis and design of RC structures.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Explain** the different methods of concrete structure design and their advantages.
2. **Analyze** and**design** singly and doubly reinforced beams, flanged beams, slabs, and staircases using the Limit State Method.
3. **Evaluate** the behavior of RC members under shear, torsion, and combined loading conditions.
4. **Design** short columns and footings considering axial and eccentric loading conditions.
5. **Utilize** IS code provisions and design aids for efficient structural design.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**
CO -1	**3**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**3**	**2**
CO -2	**3**	**3**	**3**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**3**	**2**
CO -3	**3**	**2**	**2**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**2**	**2**
CO -4	**3**	**3**	**3**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**3**	**3**
CO -5	**2**	**2**	**2**	**1**	**3**	**-**	**-**	**-**	**-**	**2**	**1**	**2**	**2**	**2**

UNIT – I

METHODS of DESIGN of CONCRETE STRUCTURES

Concept of Elastic Method, Ultimate Load Method and Limit State Method – Working Stress Method As Detailed in IS Code - Design of Singly Reinforced Beam By Working Stress Method - Limit State Philosophy As Detailed in IS Code - Advantages of Limit State Method Over Other Methods - Analysis and Design of Singly and Doubly Reinforced Rectangular Beams By Limit State Method.

UNIT – II

LIMIT STATE METHOD - FLANGED BEAM, SHEAR &TORSION

Analysis and Design of Flanged Beams – Use of Design Aids for Flexure - Behaviour of RC Members in Bond and Anchorage - Design Requirements As Per Current Code - Behaviour of RC Beams in Shear and torsion - Design of RC Members for Combined Bending, Shear and torsion - Serviceability.

UNIT – III

LIMIT STATE DESIGN of SLABS and STAIRCASE

Analysis and Design of Cantilever, One Way, Two Way and Continuous Slabs Subjected to Uniformly Distributed Load for Various Boundary Conditions- –Introduction to Flat Slab.

UNIT – IV

LIMIT STATE DESIGN of COLUMNS

Types of Columns – Design of Short Rectangular and Circular Columns for Axial, Uniaxial and Biaxial Bending.

LIMIT STATE DESIGN of FOOTING

Design of Wall Footing – Design of Axially and Eccentrically Loaded Rectangular Pad and Sloped Footings – Design of Combined Rectangular Footing for Two Columns Only.

UNIT – V

LIMIT STATE of SERVICEABILITY and MISCELLANEOUS (Aspects of Deflection, Cracking aspects) Types of Staircases – Design of Dog-Legged Staircase

TEXT BOOKS:

1. B. C. Punmia, Ashok Kumar Jain andArun Kumar Jain, Limit State Design, Laxmi Publications Pvt. Ltd., New Delhi
2. P. C. Varghese, Limit State—Designed of Reinforced Concrete ,Prentice Hall of India, New Delhi

REFRENCE BOOKS:

1. N. Krishnaraju, —Structural Design and Drawing, UniversitiesPressPvtltd, Hyderabad.4rdedition 2020.
1. N.C. Sinha and S.K. Roy,—Fundamentals of Reinforced Concrete, S. Chand Publishers
2. N.Subramanian, —Design of Reinforced Concrete Structures, Oxford University Press

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/105/105105105/

Codes/Tables: IS 456-2000 and relevant sheets (Pertaining to columns) of SP 16 Code books to be permitted into the examinations Hall.

NOTE: Assignment on preparation of drawing sheets detailing various RC Elements

All the designs to be taught in Limit State

Method Following plates should be prepared by the students.

1. Reinforcement particulars of T-beams and L-beams.
2. Reinforcement detailing of continuous beams.
3. Reinforcement particulars of columns and footings.
4. Detailing of One-way, Two way and continuous slabs

Exam Pattern:

The end examination paper should consist of Part A and Part B.

Part A consists of two questions in Design and Drawing out of which one question is to be answered.

Part-B should consist of five questions on design out of which three are to be answered.

Weightage for Part -A is 40% and Part-B is 60%.